Answer Key — Day 2: Solving Practice

Part 1: Multi-Step Isolation

1. 31: $3\sqrt[3]{x-4} = 9 \to \sqrt[3]{x-4} = 3 \to x-4 = 27$.

2. 108.5: $2\sqrt[3]{2x-1} = 12 \to \sqrt[3]{2x-1} = 6 \to 2x-1 = 216$.

3. 218: $3(x-2)^{1/3} = 18 \to (x-2)^{1/3} = 6 \to x-2 = 216$.

4. 27: $-2\sqrt[3]{x} = -6 \to \sqrt[3]{x} = 3 \to x = 3^3$.

Part 2: Radicals on Both Sides

5. 1: $8(2x-1) = 3x+5 \to 16x-8 = 3x+5 \to 13x = 13$.

6. 2, -2: $x^2+2x-5 = 2x-1 \to x^2 = 4$.

7. $\pm$1.34: $15x^2-19 = x^2+6 \to 14x^2 = 25 \to x^2 = \frac{25}{14}$.

8. 3: $2x+3 = 5x-6 \to 9 = 3x$.

Part 3: Rational Exponents ($\pm$)

9. 64, -64: $x = \pm 16^{3/2} = \pm 4^3$.

10. 9, -7: $x-1 = \pm 4^{3/2} = \pm 8$.

11. -3, -5: $x+4 = \pm 1^{3/2} = \pm 1$.

12. 9, -7: $x-1 = \pm 16^{3/4} = \pm 2^3$.

Part 4: Advanced Factoring

13. 9, 7: $x^2-16x-1 = -64 \to x^2-16x+63 = 0 \to (x-9)(x-7) = 0$.

14. 2, 3: $p^2-5p-58 = -64 \to p^2-5p+6 = 0 \to (p-2)(p-3) = 0$.

15. 7, -4/7: $7x^2-42x-1 = 3x+27 \to 7x^2-45x-28 = 0 \to (7x+4)(x-7) = 0$.

16. 6, -3: $8(5p^2) = 8p+144 \to 40p^2-8p-144 = 0 \to 5p^2-p-18 = 0$.

Part 5: Challenge & Non-Factorable

17. -4, -1, 2: $3x+8 = (x+2)^3 \to 3x+8 = x^3+6x^2+12x+8 \to x^3+6x^2+9x = 0 \to x(x+3)^2 = 0$.

18. 0, 2, -2: $4x = x^3 \to x^3-4x = 0 \to x(x^2-4) = 0$.

19. $2 \pm \sqrt{2}$: $x^2-2x+1 = 2x-1 \to x^2-4x+2 = 0$. Use Quadratic Formula.

20. 1.37, -0.81: $9x^2-7x+2 = -2x+12 \to 9x^2-5x-10 = 0$. Use Quadratic Formula.

Part 6: Applications

21. 125: $\sqrt[3]{r} = 0.5(10) = 5 \to r = 5^3$.

22. 3.4: $\sqrt[3]{L} = 0.1(15) = 1.5 \to L = 1.5^3 = 3.375$.

You Try!

1. $\pm$3: $x^2-1 = 8 \to x^2 = 9$.

2. 115, -135: $x+10 = \pm 125$.

3. 256: $x = 128^{8/7} = (2^7)^{8/7} = 2^8$.

4. 9, -2: $x^2-7x-10 = 8 \to x^2-7x-18 = 0 \to (x-9)(x+2) = 0$.